

1 Introduction

I am a number theorist with a focus on Diophantine approximation. This means that I study questions about the integers by proving relevant inequalities. Oftentimes, those inequalities relate to questions about how close rational numbers can be to irrational numbers.

For example, the integral solutions (x, y) to Pell's equation $X^2 - 2Y^2 = 1$ correspond to rational numbers x/y which are close to $\sqrt{2}$. To see why this is the case, first note that it suffices to find the solutions where x and y are nonnegative since any time (x, y) is a solution, $(\pm x, \pm y)$ will also be a solution. Second, there is only one such solution with $y = 0$, namely $(x, y) = (1, 0)$, so we may also assume $y \neq 0$. Next, divide both sides of Pell's equation by Y^2 and factor the left-hand side to find that

$$\left(\frac{X}{Y} - \sqrt{2}\right)\left(\frac{X}{Y} + \sqrt{2}\right) = \frac{1}{Y^2}.$$

For an integral solution (x, y) with $y \neq 0$, the right-hand side will be small, implying that one of the two factors on the left-hand side must be small. Since x and y are positive, then it must be $x/y - \sqrt{2}$ that is small, so x/y is a good approximation of $\sqrt{2}$. Since x/y is rational, every positive solution to Pell's equation produces a good rational approximation of $\sqrt{2}$. Hence, facts about rational approximations of $\sqrt{2}$ can be used to solve Pell's equation.

My work in this field can be separated into two broad categories:

1. Use a similar approach to solve other types of equations known as Thue equations.
2. Develop new inequalities which can be used when solving Diophantine equations.

2 Thue Equations

A **Thue equation** is an equation of the form $F(X, Y) = h$ where h is an integer and $F(X, Y)$ is a polynomial of the form

$$F(X, Y) = \sum_{j=0}^n a_j X^{n-j} Y^j$$

where each a_j is an integer, the degree of F is at least 3, and $F(X, Y)$ is irreducible over $\mathbb{Q}[X, Y]$. These equations lend themselves nicely to the same analysis as Pell's equation, but with one key difference: there are only finitely many integral solutions to Thue equations [7], while there are infinitely many solutions to Pell's equation [6].

Since there are only finitely many integral solutions to Thue equations, my work focuses on bounding the number of integral solutions. In my paper [4], I improve the upper bounds on the number of solutions to Thue equations of the form $aX^n + bX^k Y^{n-k} + cY^n = \pm 1$. In my dissertation [3], I improve the upper bounds on the number of solutions to more general Thue equations. I also have ongoing work with Eva Goedhart and Sumin Leem concerning Thue equations where the solutions and coefficients are allowed to come from more general families of "integers" (rings of integers from imaginary quadratic number fields).

In the future, I would like to extend my work with Eva and Sumin from the degree 4 case (which we are currently considering) to the degree 6 case. Additionally, I would like to extend my work on Thue equations of the form $aX^n + bX^kY^{n-k} + cY^n = \pm 1$ to Thue equations of the form $aX^n + bX^kY^{n-k} + cY^n = h$ for any integer h .

A potential undergraduate research project concerning Thue equations would be updating the results from [2] to account for new parameters that are used in modern analysis around Thue equations. For instance, many bounds on the number of solutions to Thue equations are now expressed in terms of the number of summands of $F(X, Y)$, whereas Bombieri and Schmidt use the degree of $F(X, Y)$ in [2]. Updating Bombieri and Schmidt's results to account for the number of summands of $F(X, Y)$ would be a good undergraduate project: a student can use the same approach as Bombieri and Schmidt, while making modifications to their proofs along the way. This project would work best for summer research because it requires a bit more immersion than the project I'll describe in the next section.

3 Diophantine Inequalities

My other major research category is that of developing new inequalities which can be used for Diophantine equations. As in the Pell's equation example, we can learn about solutions to Diophantine equations through our understanding of the roots of polynomials (in that example, we learned about solutions to $X^2 - 2Y^2 = 1$ through understanding $\sqrt{2}$).

One key feature to understand (for the purposes of Diophantine approximation) is how the roots of a polynomial are distributed in the complex plane. My work with Seda Albayrak, Samprit Ghosh, and Khoa Nguyen in [1] gives information about the distribution of absolute values of roots of polynomials by giving lower bounds on certain products of the form $|\alpha_0||\alpha_1|^{c_1} \cdots |\alpha_k|^{c_k}$ where the α_i are algebraic integer conjugates listed in descending order of absolute value. My work with Chi Hoi Yip in [5] gives information about the distances between roots of a polynomial in terms of information about its coefficients.

This latter project is especially well suited for undergraduate research. Given a polynomial $f(x)$, you can ask how far apart its roots can be (or how close they can get). Chi Hoi and I prove an inequality relating two quantities: one is called the root separation, which contains information about the roots of the polynomial, and one is called the Mahler measure, which contains information about the coefficients of the polynomial. While we find an optimal exponent in that inequality, we do not find an optimal coefficient. In certain cases, however, finding an optimal coefficient can be a straightforward process: first, use geometry and calculus to find an "optimal" configuration for the roots of the polynomial (e.g. the vertices of a square, or equilateral triangle). Second, do a calculation relating the separation and Mahler measure for that optimal configuration.

This type of project would only require a background in calculus, an introductory proofs course, and familiarity with the complex numbers. As a result, it would work well in many different settings where undergraduate research can take place.

References

- [1] Seda Albayrak, Samprit Ghosh, Greg Knapp, and Khoa D. Nguyen. On certain polytopes associated to products of algebraic integer conjugates. Accepted for publication in the *Proceedings of the American Mathematical Society*, preprint available at <https://arxiv.org/abs/2408.00250>, 2024.
- [2] Enrico Bombieri and Wolfgang M. Schmidt. On Thue’s equation. *Invent. Math.*, 88(1):69–81, 1987.
- [3] Greg Knapp. *Polynomial Root Distribution and Its Impact on Solutions to Thue Equations*. PhD thesis, University of Oregon, 2023.
- [4] Greg Knapp. The number of solutions to the trinomial Thue equation. *Functiones et Approximatio Commentarii Mathematici*, 69(2):247 – 270, 2023.
- [5] Greg Knapp and Chi Hoi Yip. Upper bounds on polynomial root separation. *Bulletin of the Australian Mathematical Society*, page 1–16, 2025.
- [6] Kenneth H. Rosen. *Elementary number theory and its applications*. Addison-Wesley, Reading, MA, fourth edition, 2000.
- [7] Axel Thue. Über Annäherungswerte algebraischer Zahlen. *J. Reine Agnew. Math*, 135(135):284–305, 1909.